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## The Data Science Era's Tour: A Statistical Journey Through Taylor Swift's Discography

### Introduction

According to Spotify, Taylor Swift was the top global artist of 2024, with more than 26.6 billion streams globally (“Revealed: The Top Artists,” 2024). The influence that Taylor Swift has had on music is undeniable and understanding her music may be important for current and future musical artists. Additionally, data about her music could be interesting to her fans around the world. In this paper, a dataset containing much of Taylor Swift's discography is analyzed for trends, patterns, and characteristics of her music. This dataset was found from Kaggle and was created by Yamini Manral and Rushikesh Karwankar (Manral and Karwankar, 2024). This dataset includes all songs and albums from the time of the release of 1989 to present day. This dataset also includes information about Swift's tracks including run time, track number, explicitness, and release date. Additionally, it includes characteristics about the danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, and valence of each track (Manral and Karwankar, 2024).

While this dataset did not clearly define some of these items, other similar datasets with the same values have. Danceability represents how suitable a track is for dancing based on musical elements including tempo, rhythm stability, beat strength, and overall regularity, where a value of 0.0 is least danceable and 1.0 is most danceable (Priester, 2024). Energy is a measure from 0.0 to 1.0, which represents a measure of intensity and activity in the music, where more

fast, loud, and noisy tracks rank closer to 1.0 (Priester, 2024). Loudness is the overall loudness of the track in decibels (dB) (Priester, 2024). Speechiness detects the presence of spoken words in a track, so 1.0 would be all words and 0.0 would be all music (Priester, 2024). Acousticness represents a measure from 0.0 to 1.0 of whether the track is acoustic or not where 1.0 represents high confidence that the track is acoustic (Priester, 2024). Instrumentalness measures whether a track contains no vocals, so 1.0 represents a track with no vocals and closer to 0.0 represents a track with a lot of vocals (Priester, 2024). Liveness detects the presence of an audience in the recording, where the closer to 1.0 the ranking is, the more likely that there is an audience (Priester, 2024). Finally, valence is a measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track (which invoke feelings of happiness and cheerfulness), where a more positive sounding track will have a high valence score (Priester, 2024). Understanding these factors could be important for artists to determine what will be popular among worldwide audiences and this data could potentially help them create popular music.

This data can answer those important questions for artists and fans of Taylor Swift. For this paper, the following questions will be explored to better understand Swift's music:

- What does a timeline for all the Taylor Swift albums listed in this dataset look like?
- Which Taylor Swift album has the highest average danceability according to this data?
- Are danceability and energy of Taylor Swift songs correlated?
- Is there a correlation between explicitness of Taylor Swift tracks and the album they're found on?
- Is there a connection between the explicitness of Taylor Swift songs and their characteristics (such as danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, and valence)?

- What are the distributions of acousticness shown in each album?

By answering these questions, fans of Taylor Swift and other artists looking to make successful music like Taylor Swift can understand her music better. Additionally, this not only shows trends in this data, but can visualize these and other unique parts about her music for any audience.

## Methods

To answer questions about this dataset, Excel and Google Colab were used. The data from Google Colab can easily be accessible in Python and uses the same methods as that programming language. The first question that will be answered in this paper is “What does a timeline for all the Taylor Swift albums listed in this dataset look like?” The different ways to format this data were generated using OpenAI’s *ChatGPT* (OpenAI, 2025). These simply provided examples of how timelines could be created using a general dataset. To answer the question for the Taylor Swift dataset, the Pandas library was used in Google Colab to create a new data frame based on the information from the Taylor Swift dataset (OpenAI, 2025). This included the album name and the release data of that album. Then, using the Matplotlib library, the release dates of each album included in the dataset were organized in order (OpenAI, 2025). These were plotted using a colorful palette from this library and album names were chosen to be displayed on the y-axis, while the release date was chosen to be represented on the x-axis (OpenAI, 2025).

The second question that was posed for this data was “Which Taylor Swift album has the highest average danceability according to this data?” This question was answered using Excel. By importing all data into Excel and selecting all of it, a PivotTable was inserted using the

PivotTable creation tool under the “Insert” tab. The columns of the PivotTable were selected to be the album names and the values were selected to be the average danceability. For here, a bar chart was created to show the corresponding average danceability values in a figure.

The next question to be answered was “Are danceability and energy of Taylor Swift songs correlated?” Google Colab was used to answer this question. Pearson’s correlation score was used to determine if these were correlated. This was done using the Scipy library’s statistics package, where a result close to 0 represents no correlation, a result close to 1 represents a strong, positive correlation, and a result close to -1 represents a strong, negative correlation. Additionally, this result was visualized using a linear regression model completed using the Scipy statistics package. This method determines the slope and y-intercept of a line of best fit, which was added onto a scatterplot with danceability against energy created using Matplotlib. This helps show how correlated these variables are.

Next, the following question was explored: “Is there a correlation between explicitness of Taylor Swift tracks and the album they're found on?” To answer this question, Google Colab was used. Using the Pandas library, a cross tab was created to understand the relationship between explicitness and album name. Then, using the Scipy stats package, a Chi Square analysis was run using the Chi Square contingency test in the library. This resulted in a “p” value to represent this relationship. If the value was below 0.05, then this represents a significant relationship between the album and whether a song is explicit or not.

The next question to be explored was “Is there a connection between the explicitness of Taylor Swift songs and their characteristics (such as danceability, energy, loudness, speechiness, acousticness, instrumentality, liveness, and valence)?” Google Colab was used to answer this question and machine learning was heavily utilized. To begin, the explicit column was changed

from booleans to integers using an `.astype(int)` function. Then, variables for X data (the traits listed above) and y data (explicitness) were created and the train-test split was performed on these variables. Then, the classifier was defined with a certain number of neighbors and go through the steps to perform the k Nearest Neighbors function. This will help make predictions about the data. Additionally, an accuracy score was reported, and a confusion matrix was created to show the accuracy of the model in a visualized format. From there, the model was improved using hyperparameter tuning (changing the number of nearest neighbors) and feature engineering (changing the traits used in the X data). This method helps to determine how connected these variables are using a machine learning algorithm and determines how accurately it can model this relationship.

Finally, the last question explored was: “What are the distributions of acousticness shown in each album?” To answer this question, the Joyplot library was used in Google Colab (Shah, 2024). This library helped to create a ridgeline chart with only a few lines of code. This type of chart will show the distribution of the data for a certain column used (Shah, 2024). In this case, this was done by organizing the data by the album name using the acousticness column (Shah, 2024). This chart was displayed using the Matplotlib library to show the distributions of acousticness for each album.

## Results

The results of the tests and methods performed for this dataset are shown below in Figure 1, Figure 2, Figure 3, Figure 4, and Figure 5. A timeline of all Taylor Swift albums releases is shown in Figure 1. The average danceability of each album can be seen in Figure 2. The correlation between danceability and energy can be seen in a scatterplot in Figure 3. The results

of the machine learning algorithm used to determine if there a connection between the explicitness of Taylor Swift songs and their characteristics are shown in two confusion matrixes in Figure 4. Finally, Figure 5 shows the distribution of acousticness amount the albums included in this dataset. Additionally, the Chi Square test showing the if explicitness is correlated to what album the song is found on resulted in a p-value of  $3.866 \times 10^{-12}$ .

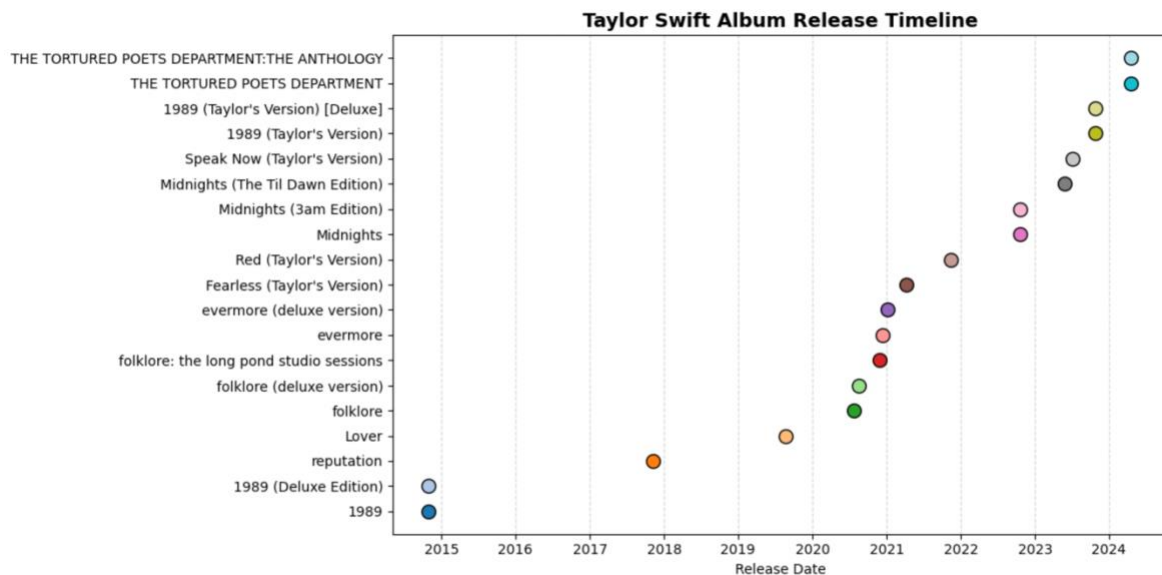


Figure 1: Timeline showing All the Albums of Taylor Swift included in this Data Set: This timeline shows all of Swift's albums listed in timeline order from 1989 to THE TORTURED POETS DEPARTMENT: THE ANTHOLOGY. This shows that Swift has had most of her releases over the past 5 years.

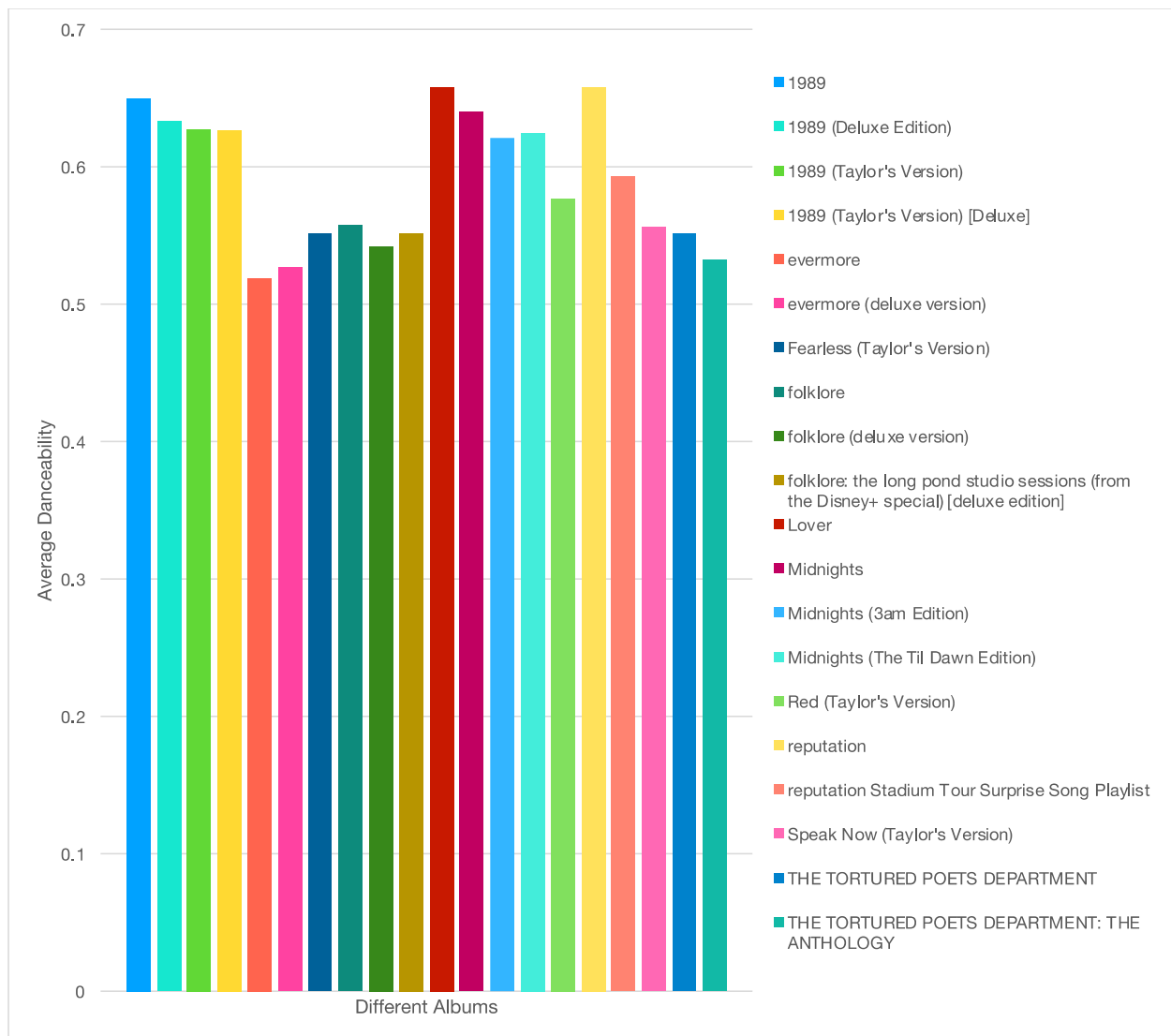


Figure 2: A Bar Chart of the Average Danceability Ranking of Each Taylor Swift Album: This bar chart shows the average danceability of each album in the dataset. This shows that Lover has the highest average danceability and evermore has the lowest average danceability.

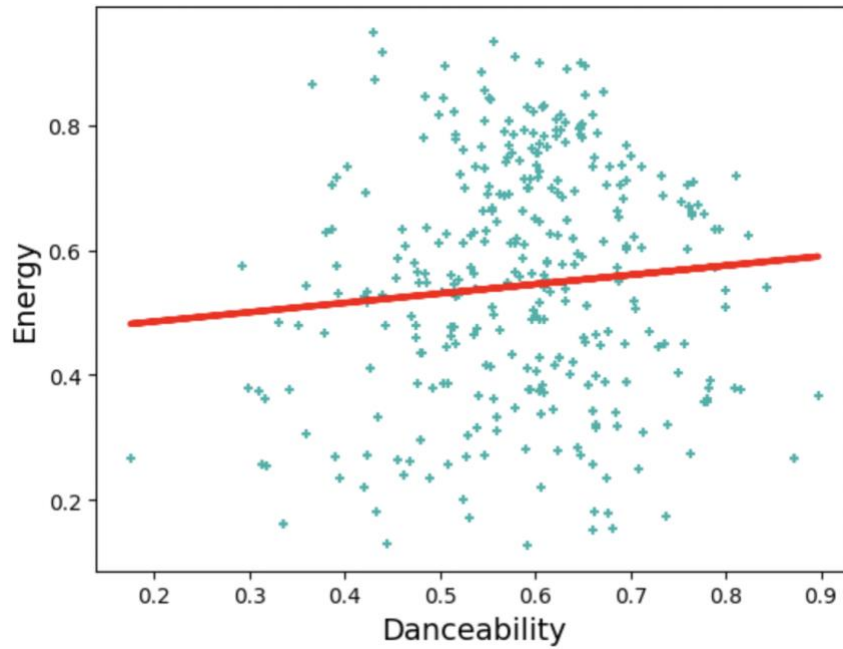


Figure 3: A Scatterplot of Danceability and Energy with a Line of Best Fit: The scatterplot shows danceability and energy plotted against each other to determine if they are related or not. The line of best fit was created using the Scipy library and showed a Pearson R value of 0.0919, showing that these characteristics are not related in Taylor Swift's songs.



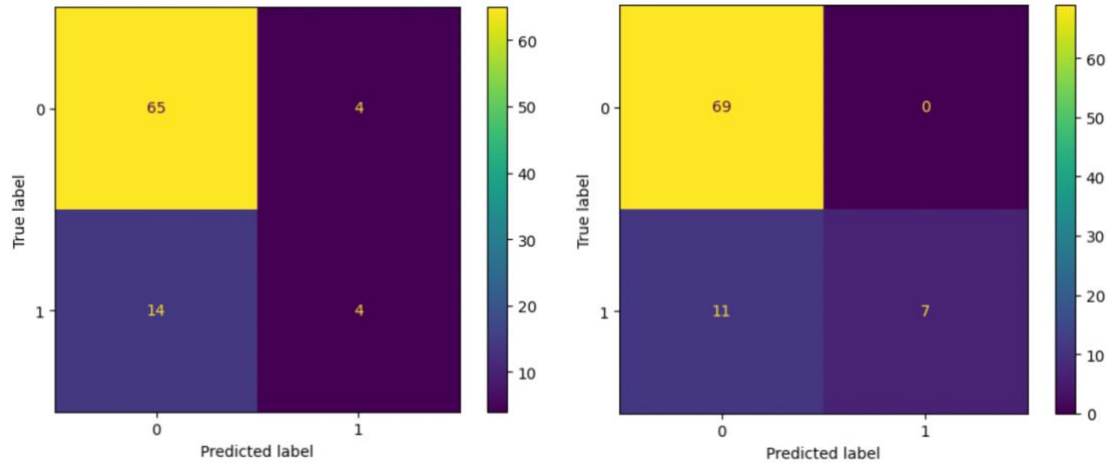


Figure 4: Confusion Matrixes for the Machine Learning algorithm that tried to Predict the Explicitness of a Taylor Swift Song based on the Traits of the Song (including danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, and valence): The confusion matrix on the left shows the result of the first algorithm tested where 65 were predicted to not be explicit and were not explicit, 14 were predicted to not be explicit but were not actually explicit, 4 were predicted to be explicit but were actually explicit, and 4 were predicted to be explicit and were explicit. This is an overall accuracy of 79.31%. To improve this accuracy, hyperparameter tuning was performed and instead of using 5 neighbors, 2 neighbors were used. The confusion matrix on the right shows the results after this adjustment and the accuracy improved. This is shown here as now 69 were predicted to not be explicit and were not explicit, 11 were predicted to not be explicit but were not actually explicit, 0 were predicted to be explicit but were actually explicit, and 7 were predicted to be explicit and were explicit. This improved the overall accuracy to 87.36%.

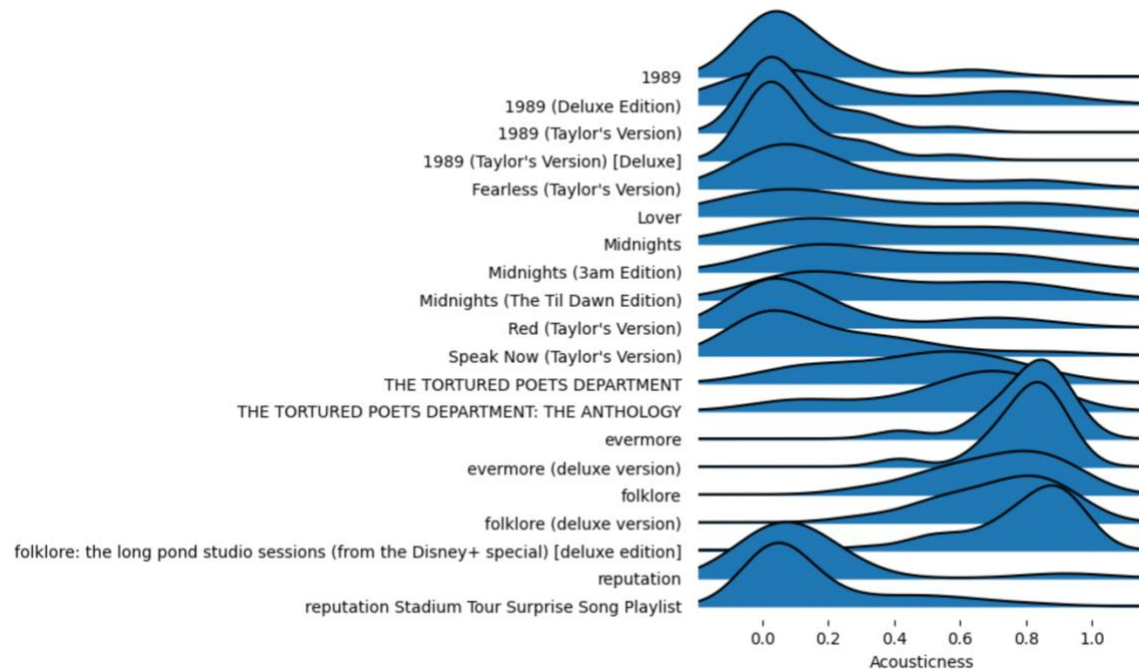


Figure 5: Ridgeline Chart showing the Distribution of Acousticness in Each Taylor Swift Album: This chart helped display how acousticness was found in each album in the dataset. This shows that the most acoustic albums were the different versions of folklore and evermore, while the versions of 1989 and reputation seem to show the least amount of acousticness.

## Discussion

From this data, many conclusions can be drawn about Taylor Swift songs, albums, and the characteristics of both. When looking at a timeline of the album releases using this dataset, it seems like only the music Taylor Swift made after the original version of 1989 were included (Figure 1). This means that for the purposes of this data, her first four albums (Taylor Swift, Fearless, Speak Now, and Red) were not included. Based on the timeline of the albums in this dataset, it seems like most of Swift's album releases have been within the last five years with her last album release being just over a year ago (Figure 1).

While not all albums were included in this dataset, it is important to look at the trends and statistics of those that are included. As stated before, danceability represents how suitable a track is for dancing based on musical elements including tempo, rhythm stability, beat strength, and overall regularity, where a value of 0.0 is least danceable and 1.0 is most danceable (Priester, 2024). This statistic can be important to listeners, as a more danceable album could attract new listeners and demographics. When looking at this factor in Taylor Swift albums, *Lover* shows the highest average danceability based on its songs (Figure 2). Additionally, all versions of *1989*, *Midnights*, and *reputation* show high average danceability (Figure 2). These albums are all categorized as the Pop genre of music, which is known for its popularity with a wide range of audiences due to its catchy rhythms, upbeat nature, and danceability (*Damvibes*, 2018). This could be why a lot of Swift's music has been in this genre, as it attracts more people to listen to her music.

It is also important to see how characteristics of Taylor Swift songs are related to determine if other traits impact each other and how this could impact the music for audiences. While there are many characteristics observed in this dataset, one of the ones that was interesting was the correlation between danceability and energy since these can often be related when discussing music. The relationship between these factors showed a Pearson R value of 0.0919, which is a value that shows a light to no correlation between these factors (Figure 3). This is shown in the line of best fit in Figure 3 as well. This shows that the data between these variables are very spread out and do not show a correlation (Figure 3). This means that artists who want to make more popular music like Taylor Swift's music can make danceable music without high energy/intensity or vice versa.

In addition to correlations in numerical data, it is also important to look at and interpret the relationships between categorical data. In this case, explicitness and the album a track was on were used to see if there was a significant relationship observed there. This was done using the Chi Square test, which resulted in a p-value of  $3.866 \times 10^{-12}$ . This shows a very low p-value, indicating that the explicitness of a track is related to the album it is found on. This could be important to observe, as this could be an intentional decision by Swift to use explicit language to evoke the feeling of an album. For example, 1989 (Taylor's Version) has 21 songs and none of them are explicit, while THE TORTURED POETS DEPARTMENT has 16 songs and 9 of them are explicit. 1989 (Taylor's Version) is a very upbeat and playful pop album, while the messaging of THE TORTURED POETS DEPARTMENT is much darker. Using more intense language seems to be a storytelling strategy that Swift uses for these albums to give different feelings to the overall album. This is shown by this Chi Square relationship and through listening to her different albums.

These characteristics may also show relationships to other factors within the music as well. It was of interest to see if certain characteristics could predict other characteristics about a Taylor Swift song. In this case, the characteristics of danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, and valence could help predict if a track would be explicit or not. The k Nearest Neighbors algorithm used was able to predict explicitness of a track based on these characteristics was about 79.31% accurate (Figure 4). This was using 5 neighbors in the k Nearest Neighbors algorithm, but 2 neighbors was found to be best when using hyperparameter tuning. When this was adjusted, the accuracy increased to 87.36% (Figure 4). This shows that using these variables, explicitness can be predicted pretty accurately. This is important for the music industry, as once again this shows that explicit language could be used

strategically to adjust the feel of a song or album. This can have an impact on the audience as well, which is useful to know as a musical artist.

In addition, it was interesting to see what the distribution of a certain characteristic are seen in each of Taylor Swift's albums. One trait that has not been looked at directly yet was acousticness, which represents a measure from 0.0 to 1.0 of whether the track is acoustic or not where 1.0 represents high confidence that the track is acoustic (Priester, 2024). Looking at the distribution of that throughout Taylor Swift albums could be telling on how well these albums sell or connect with audiences. The distributions of acousticness in each of the albums listed in this dataset are shown in Figure 5. The different versions of *folklore* and *evermore* seem to show more acousticness throughout the album's tracks (Figure 5). As of April 2024, *1989*, *Red*, *Lover*, *Midnights*, *reputation*, and *Fearless* all ranked higher than *folklore* and *evermore* in total album sales (*MusicWeek*, 2024). This could mean that less acoustic albums sell better in the long run, and this could impact the music that people see being produced in the industry.

These observations made using different data analysis techniques allow listeners of Taylor Swift to have a better understanding of her music and the trends that can be observed throughout her discography. Looking at these trends could be important to musical artists who wish to make similar music to Swift and want to understand her music more to do so. Additionally, these observations could be important to understand popular music today and how well certain musical choices impact audiences and sales.

## Conclusion

The observations and analysis of this dataset have revealed various trends and relationships within Taylor Swift's music. By examining her albums from 1989 onward, many

key factors, including danceability, energy, and acousticness, were explored and able to be compared and better understood. The analysis in this paper revealed that Swift's album, *Lover*, has a high average danceability, aligning with the upbeat nature of its pop music. Additionally, there was a lack of correlation found between energy and danceability, suggesting that Swift's music can be higher in one than the other. Additionally, acousticness was found to be highest in *folklore* and *evermore* and lower in Swift's pop music.

A Chi Square test between explicitness and album name revealed that explicitness of Swift's music is dependent on the album it is found on. This shows that Swift may use explicit language intentionally to evoke the feeling of an album. In addition to this, the use of machine learning showed a high accuracy when characteristics of Swift's music were used to predict explicitness. This suggests that there is a strategic use of language and musical characteristics by Taylor Swift to get a certain feel within her songs and albums.

Overall, this analysis helps deepen the understanding of Taylor Swift's music, which can be helpful for other artists in the music industry who want to make similar music. Using data science to examine music is crucial for our understanding of the music industry and its successful artists. As music continues to evolve and more popular artists are analyzed, data science can continue to play a role in understanding artists' success and understanding the strategies used for musical success all around the world.

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